



اُنِيْوَرْسِيْٓتِيْ تِكْنُوْلُوْجِيْ مَآرَا
UNIVERSITI
TEKNOLOGI
MARA

Fakulti
Sains Komputer
dan Matematik

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES
BACHELOR OF COMPUTER SCIENCE (HONS.)

ICT502
DATABASE ENGINEERING

PROJECT PROPOSAL TITLE :

Cikgu

Personalized Learning Platform

GROUP :

NBCS2306A - Group 3

PREPARED BY:

Name	Student ID
Raef Luqman Bin Mohammad Aidid	2022673128
Amrina Binti Shamsudin	2023280798
Nurathika Nazira Binti Musa	2024478332
Siti Syazila Binti Bidin	2023954993

PREPARED FOR:

DR. NOR DIANA BINTI AHMAD

Members

Name	Profile Picture
Raef Luqman Bin Mohammad Aidid ?	
Amrina Binti Shamsudin !	
Nurathika Nazira Binti Musa ?	
Siti Syazila Binti Bidin ?	

add roles and responsibilities!!!

TABLE OF CONTENTS

1.0 Company Background.....	3
2.0 Problem Statement And Motivation.....	4
2.1 Problem Statement.....	4
2.2 Motivation.....	5
3.0 Objectives And Scope.....	6
3.1 Objectives.....	6
3.2 Scope.....	6
3.3 Objective-to-Feature Linkage.....	7
4.0 Entity Relationship Diagram.....	8
4.1 Business Rules.....	9
5.0 3NF Relational Schema.....	10



1.0 Company Background

Cikgu, a startup in the field of Ed Tech, was established in 2026 and is based in Selangor, Malaysia. The name embodies the key principle, educators at the heart of learning. In a world of online learning platforms, most have content structure that is centred around subjects, not learning outcomes.

Students have access to tutors through the web on Cikgu's customizable and goal-centric learning process. Rather than learners having to follow a myriad of unrelated courses in different subject areas, Cikgu provides learners with the opportunity to define the goal for learning, such as a job in data engineering, a career change to UX design, learning a new field, and more. It then creates an individual learning journey that is based on their prior learning and background.

Audiences encompass school and university students as well as reskilling or career switching working people. Tutors on the platform are saved from administrative tasks and burden of designing curriculum, and can focus on what they do best, teaching.

Cikgu: where teachers teach, and learners grow.

2.0 Problem Statement And Motivation

2.1 Problem Statement

Even most independent tutors and small learning centers are still on the manual or file-based system with spreadsheets, paper records, whatsapp messages and individual documents to record their learners' data. There is no central database system and during the management process there are a number of things that could occur.

Inconsistent learner records and data redundancy

Tutor, learner and enrolment information is held on a number of files and devices. Information about the same learner is duplicated, inconsistent and is out of date in 3 or 4 places. The more learners that join, the more complex and inefficient it is to keep these records by hand.

Inefficient progress tracking

Manual tracking of each learner's progress is time consuming. Hours and hours are spent by tutors updating spreadsheets, scoring, and generating reports, time that would better be spent teaching. These tasks can be time-consuming and error-prone if done manually.

No personalization across learners

All learners are unique in their goals and learning capacity. But for the most part, all learning centres are offering exactly the same learning materials to all learners as there is no appropriate system in place for managing learner information. For instance, if a learner wishes to be a Data Engineer, they might have to follow a different learning path. If not provided with personalized learning support, the learning experience may be less effective and may negatively impact learner motivation.

Difficulty in monitoring and reporting

Another issue is the inability to report and keep track of the learner's performance. When all data is spread over files, the tutors are not easily able to answer simple questions such as, which learner is behind schedule? Or what are the top performing modules? Slow and error-prone manual aggregation is needed for 'on-demand' reports.

2.2 Motivation

The trend of outcome based learning is underway but many tutoring businesses are still using manual file management. It is clear that there is a need for a structured and centralized database system that can be personalized on a large scale, that will not add administrative burden to tutors, and that will allow for monitoring of learner progress through data analysis. Hence, it is proposed to use Cikgu to facilitate the management of the learners. All the records can be maintained in a single system rather than in a number of different files. This can save tutors time, as well as minimize manual work and check learner progress more efficiently.



1. to identify system requirements for Cikgu: Personalized Learning Platform
2. to design an application and database for Cikgu: Personalized Learning Platform
3. to develop a Cikgu: Personalized Learning Platform

3.0 Objectives And Scope

3.1 Objectives

Development Objective 1 (DO1): To design a ~~normalized relational database (3NF) with at least 6 entities which includes inheritance, recursive and bridge structures to store user, learner, tutor, goal, module, and enrollment data with referential integrity.~~

Development Objective 2 (DO2): To ~~build a web-based application which will enable learners to register goals and enroll in modules as well as enable tutors to track learner progress using the SQL-based report.~~

Evaluation Objective 1 (EO1): ~~Test the effectiveness of the system when it comes to data integrity and query performance and the acceptance threshold is defined in Section 3.~~

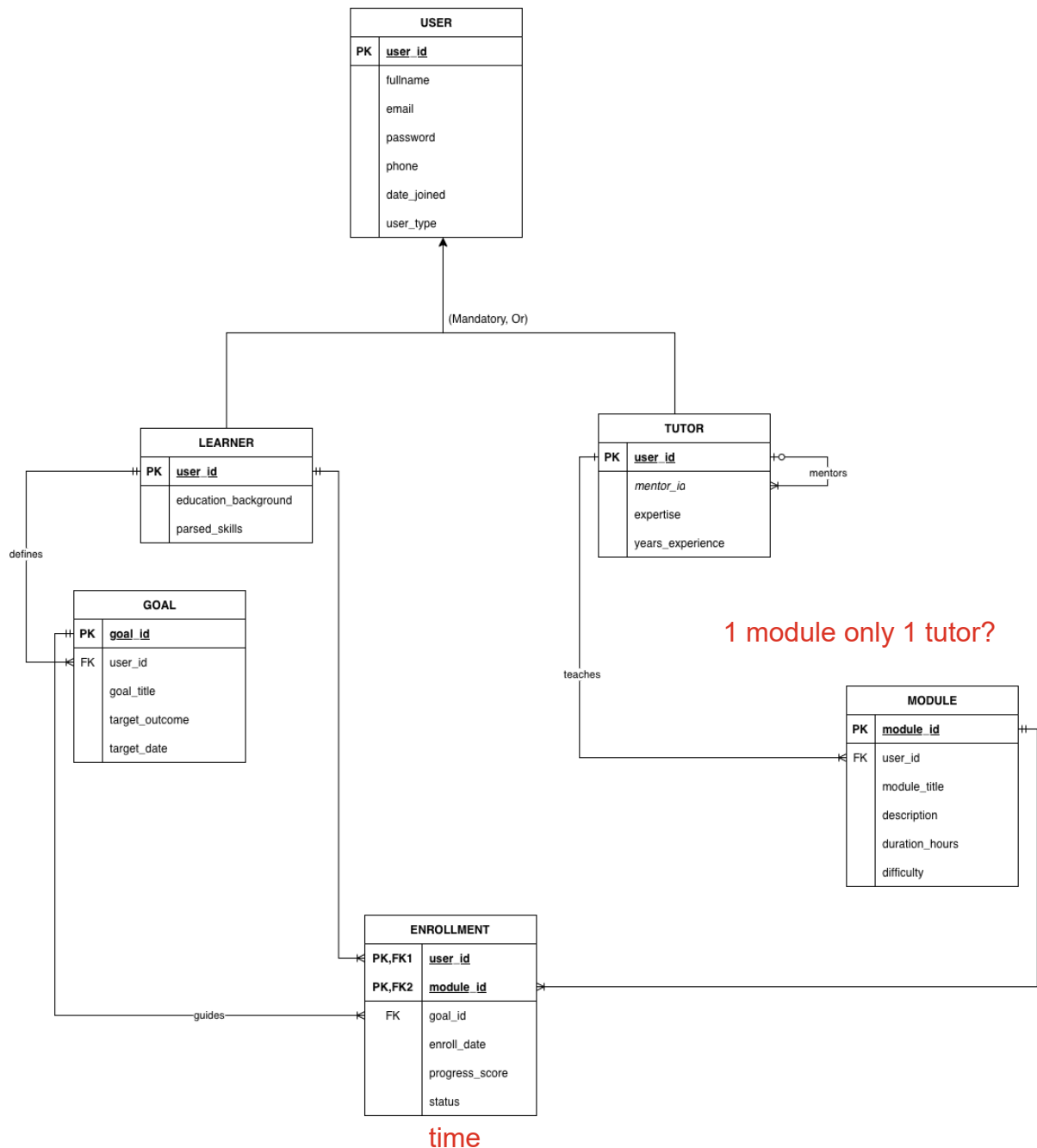
3.2 Scope

In Scope	Out of Scope
Web-based platform for learners and tutors	Payment and billing system
User registration with role-based access (Learner / Tutor)	Live video conferencing
Learner profile with education background and skills	Mobile native applications
Tutor profile with expertise and mentor hierarchy	Full AI/agentive features (future enhancement)
Goal definition per learner	Resume parsing engine
Module creation and enrollment	Assessment scoring engine
Progress tracking via enrollment status and score	Tutor session scheduling
SQL-based reports and minimum 10 ad-hoc queries	Production deployment and hosting

3.3 Objective-to-Feature Linkage

Objective	System Features
DO1	Data modeling concepts (schema, ERD, 3NF tables, integrity constraint, inheritance, recursive)
DO2	Registration of users, specification of goals, entry of modules, reporting dashboard for tutors
EO1	Integrity tests, query performance benchmarks

4.0 Entity Relationship Diagram



The Cikgu ERD consists of **6 entities** which have the following design highlights:

- **Inheritance (Specialization):** **USER** is the superclass; **LEARNER** and **TUTOR** are subclasses (disjoint specialization based on `user_type`)
- **Recursive Relationship:** **TUTOR.mentor_id** references **TUTOR.user_id**, senior tutors mentor junior tutors

- **Bridge Entity: ENROLLMENT** resolves the M:N relationship between **LEARNER** and **MODULE**, with composite primary key (**learner.user_id**, **module_id**) and **goal_id** as a non-key FK

4.1 Business Rules

The business rules below describe the operational requirements and constraints of the proposed system and were established based on the system requirements analysis. These rules are important to ensure accurate data management and consistent database operations:

- A USER is either a LEARNER or a TUTOR, determined by the **user_type** attribute.
- A LEARNER can define many GOALS.
Each GOAL belongs to one LEARNER.
- A TUTOR can mentor many other TUTORS
Each TUTOR has at most one mentor (recursive).
- A TUTOR can teach many MODULEs
Each MODULE is taught by one TUTOR.
- A LEARNER can enroll in many MODULEs through ENROLLMENT
Each MODULE can have many learners enrolled.
- Each ENROLLMENT is guided by one GOAL, indicating why the learner enrolled in that module.

5.0 3NF Relational Schema

The following schemas have been normalized to **Third Normal Form (3NF)** for the proposed database system.

Primary keys are underlined and bolded, while foreign keys are in *italics*:

1. **USER**
(**user_id**, fullname, email, password, phone, date_joined, user_type)
2. **LEARNER**
(**user_id**, education_background, parsed_skills)
3. **TUTOR**
(**user_id**, *mentor_id*, expertise, years_experience)
4. **GOAL**
(**goal_id**, *user_id*, goal_title, target_outcome, target_date)
5. **MODULE**
(**module_id**, *user_id*, module_title, description, duration_hours, difficulty)
6. **ENROLLMENT**
(**user_id**, **module_id**, *goal_id*, enroll_date, progress_score, status)